TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Wavelength of Light from a Discharge Tube by a Diffraction Grating

The grating equation (a + b) sinθ = nλ turns an angle measured on the spectrometer into the wavelength of a spectral line — here the yellow line of a helium discharge tube.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 15 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to measure the wavelength of a prominent line in a discharge tube's spectrum. A transmission grating of N lines per unit length has grating element (a + b) = 1/N — the spacing of its repeated unit. Light falling normally on it is diffracted into sharp principal maxima where the path difference between neighbouring slits is a whole number of wavelengths:

$$(a + b)\,\sin\theta = n\lambda \;\Rightarrow\; \lambda = \sin\theta / (nN)$$

with θ the diffraction angle of the n-th order. The spectrometer measures θ on both sides of the direct image and averages — cancelling any zero error in the grating's alignment. A discharge tube (helium in the model run) emits discrete lines; each line gives its own θ and hence its own λ.

2. Apparatus and procedure

- **Apparatus:** spectrometer adjusted as in Experiment 13, plane transmission grating (600 lines/mm in the model), discharge tube with its power supply (helium), darkened room.
- Mount the grating on the prism table, ruled surface facing the telescope side as directed, and set it for **normal incidence** (rotate the table until the reflected slit image coincides appropriately / use the standard two-step levelling and normal-incidence adjustment).
- Find the direct (zero-order) image; record the telescope reading.
- Swing the telescope to the first order on one side; set the cross-wire on the chosen line (helium yellow); record. Repeat on the other side. Half the difference of the two readings is θ.
- Repeat for the second order if bright enough, and for other lines (e.g. helium's red and blue-green lines). Compute λ for each.

3. Worked calculation (model data)

RESULT — HELIUM YELLOW LINE

Grating: 600 lines/mm → $(a + b) = 1/600$ mm $= 1.6667 \times 10^{-6}$ m.

First order: readings on the two sides differ by 41.30° → $\theta = 20.65°$; $\sin\theta = 0.3527$.

$\lambda = (a + b)\sin\theta/n = 1.6667 \times 10^{-6} \times 0.3527/1 = 5.88 \times 10^{-7}$ m = **588 nm**.

The tabulated helium yellow line is 587.6 nm — agreement within the reading precision of the divided circle. (Second order check: $\sin\theta_2 = 2\lambda/(a+b) = 0.7055$ → $\theta_2 = 44.9°$, the model's observed second-order position.)

4. Precautions

- Normal incidence must be set carefully: if the light strikes obliquely, the simple formula does not apply and the two sides give unequal angles — their disagreement is the diagnostic.
- Read lines on both sides of the zero order and average θ; this also cancels the circle's zero error.
- Set the cross-wire on the line's centre; bright lines have width, and one-sided settings bias θ.
- Do not touch the ruled surface of the grating; hold it by its edges and keep it dust-free.
- Discharge tubes run hot and at high voltage: allow cooling, never handle the tube by its capillary section while powered.
- Higher orders of one colour can overlap lower orders of another ($n\lambda$ equal): identify lines by checking that $\sin\theta$ scales with n for the same λ.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. What is the grating element, and why does the spectrum's sharpness improve with the number of illuminated lines?
2. A 600 lines/mm grating shows a line at $\theta = 15.05°$ in first order. Find λ and the likely source colour region.
3. What is the highest order in which the helium yellow line (587.6 nm) can appear with this grating?
4. Why are readings taken on both sides of the direct image?
5. How would you distinguish a second-order green line from a first-order red line appearing at the same angle?
6. The grating is replaced by one of 300 lines/mm. What happens to θ for the same line and order, and to the spectrum's spread overall?

Answer key

1. The grating element (a + b) is the width of one repeated unit — one transparent slit plus one opaque space — equal to 1/N. Sharpness improves with the number of illuminated slits because the principal maximum is the coherent sum over all of them: more slits concentrate the same light into a narrower angular width (resolving power = nN_total), so lines stand out as threads instead of bands.
2. $\lambda = 1.6667 \times 10^{-6} \times \sin 15.05° = 1.6667 \times 10^{-6} \times 0.2597 = 4.33 \times 10^{-7}$ m = 433 nm — the violet-blue region (helium has a line at 447.1 nm; 433 nm suggests checking the line identification and the θ reading — in the laboratory this is how a mis-set cross-wire announces itself).
3. The grating equation needs $\sin\theta = n\lambda/(a+b) \leq 1$: $n \leq (a+b)/\lambda = 1.6667 \times 10^{-6}/5.876 \times 10^{-7} = 2.84$, so the highest possible order is n = 2.
4. To cancel alignment zero errors: if the grating is not perfectly normal or the circle's zero is offset, one side's θ is too large and the other's too small by the same amount; the half-difference of the two readings is the true θ.
5. Track the orders: for the true line, sinθ at second order is exactly twice its first-order value. Observe the line's first-order position on both sides: the green line will appear at half the sinθ of the suspect position, the red line will not (its first order lies elsewhere). Colour appearance and the presence of companion lines of the same source confirm the identification.
6. sinθ halves for the same n and λ (the grating element doubles), so θ shrinks — for the yellow line, from 20.65° to about 10.2° in first order. The whole spectrum compresses toward the direct image: coarser gratings spread light less (lower dispersion), though more orders fit within 90°.
